

AGENTIC AI PLATFORM ENGINEER · MULTI-AGENT RUNTIMES ·
POLICY-FIRST CONTROL PLANES



Karen Kilroy

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Summary

Software engineer and architect with 20+ years of building and operating production systems, now focused entirely on agentic AI. Designed the layer that governs which tools and data six AI agents are allowed to use inside a HIPAA-regulated oncology prior authorization pipeline, together with the audit system that proves what each agent did, why it was permitted, and which model answered. Builds the whole path personally: agent orchestration, backend services and APIs, the database layer, and the React and command-line interfaces operators work in. Creator of Free2PA, which verifies signed agent instruction files before they reach a model. Six-time IBM Champion, winner of the IBM Watson Build Challenge North America, O'Reilly author of four books, and Co-Chair of the C2PA AI/ML Task Force.

Core Skills

Agentic AI

Multi-Agent Orchestration Custom Agent Harnesses & Runtimes Ports-and-Adapters Capability Layers
Typed Capability Contracts & Registries Model Context Protocol (MCP)
Function Calling & Structured Outputs Human-Approval Gates & Recovery Durable Session & Workflow State
RAG & Model Routing Event-Driven Queue-Based Dispatch

Engineering

TypeScript, Node.js, React, Next.js Python (intermediate) PostgreSQL, Prisma, Schema Design & Migrations
Multi-Tenant APIs, OpenAPI Azure AI Foundry, Azure OpenAI, Azure ML, FHIR
AWS Agent Deployment & Migration Service Bus, Queue Workers, KEDA, SSE OpenTelemetry, App Insights
CI/CD, Layered Testing, Release Gates

AgentOps & Governance

Hash-Chained Forensic Execution Traces Token & Cost Accounting per Agent Step
Model Provenance Ledgers & Drift Detection Least-Privilege Tool & Data Access
Prompt-Injection & Untrusted-Instruction Defenses Claude Code Skill Authoring & Code Generators
Content Provenance (C2PA, SMPTE/ETC) PHI-Safe Review Paths, RBAC / Managed Identity

Experience

Founder & Principal Engineer

Kilroy AI LLC

2026 – Present

- **Hidalga (client engagement)** — Architected the agent capability abstraction layer for a HIPAA-regulated six-agent oncology prior authorization pipeline: 16 domain-named capabilities over 24 swappable backends, so agents never bind to a vendor or model and backends change with zero agent code changes.
- Designed a typed capability outcome contract returning explicit unavailability rather than throwing, converting silent agent degradation into loud configuration errors, then codified the boundary as a project standard and drift-audited every agent against it.
- Converted four deterministic agents to LLM-backed agents on Azure AI Foundry for clinical extraction, document summarization, and payer narrative, while keeping every approval decision deterministic — the model reasons and writes, never renders the verdict.
- Built forensic agent observability: hash-chained tamper-evident execution traces with chain verification, liveness and stall detection, live SSE streaming, per-step token accounting, and model-attributed audit entries.
- Integrated an Azure ML prediction model behind a capability with a four-state trust contract, model-provenance ledger, and drift detection, so an unvalidated model declares itself rather than looking identical to a validated one.
- Authored 13 internal Claude Code skills including wiring code generators used as the anti-drift mechanism, and created PAM, a credential-free browser-driving QA agent that acceptance-tests user journeys and files field-level evidence.
- **RadioHead** — Built an autonomous broadcast transcription agent for an NPR member station, started on OpenClaw and migrated onto AWS for production reliability. Built with a student collaborator; won a student competition.
- **Free2PA** (free2pa.org) — Public provenance toolkit verifying signed agent control files before model context load: allow, deny, and quarantine primitives for a policy-first control plane, published as an open reference implementation.
- **Phyllis** (phyllis.bot) — Multi-tenant fulfillment API for autonomous commerce agents, with the consequential-action boundary designed in: agents prepare orders, humans approve anything that spends money or ships goods.

Founder & AI Engineer

NYX NoCode

2024 – Present

- Built and operate a platform where non-specialists describe an application in natural language and get a deployed React app — agent orchestration, model routing, RAG memory, code generation, and hosting, owned end to end.
- Serve public school customers in production, with the reliability, cost, accessibility, and safety obligations that implies.
- Ran hackathons and training programs that turned non-engineers into shipping builders.

Co-Chair, AI/ML Task Force

C2PA · contributor, SMPTE/ETC AI/ML Task Force

Current

- Co-chair cross-industry work on AI provenance, content authenticity, and standards adoption, then ship reference implementations so the standard becomes something teams can adopt.

Author

O'Reilly Media

2019 – 2024

- Four books: *Natural Language and Search* (2024), *Blockchain Tethered AI* (2023), *AI and the Law* (2021), *Blockchain as a Service* (2019), plus the O'Reilly Radar article "AI's Opaque Box Is Actually a Supply Chain."

CEO & Technical Lead

Kilroy Blockchain

2016 – 2025

- Led the team that built **RILEY**, winner of the IBM Watson Build Challenge North America (2017), an AI accessibility system for people who are blind or visually impaired.
- Directed engineering for blockchain-backed workflow and case management systems where auditability and operational control were core requirements.

Earlier Roles (Condensed)

- **CTO – Jamersan LLC** (2016) · **Principal Application Developer – CA Technologies** (2014–2015) · **Training Consultant – Magento Inc.** (2010–2014), founding member of Magento U · **Director of Online Marketing – Suarez Corp.** (2010–2012)

Recognition, Certifications & Education

IBM Watson Build Challenge Winner, North America (2017) · IBM Champion, six-time honoree (2020–2025)

AI Fluency for Students and Teaching the AI Fluency Framework — Anthropic (2025) · Venture Building — Builders + Backers (2025)

IBM Certifications: Watson Chatbot, RPA, Bluemix Essentials, Blockchain Essentials · FAA Private Pilot License

University of Arkansas, Sam M. Walton College of Business — Infrastructure & Cloud Computing (Spring 2026), 4.0 GPA; Quantum Computing (Fall 2026)

University of Arkansas — Studies in Music, 4.0 GPA (2023–Present) · **Hammel College** — Office Automation & Database Management (1981–1982), 4.0 GPA